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The State of Comon's Conjecture

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Thesis presented in fulfillment of the requirements for the degree of Master of Science in Mathematics

Academic year 2025 -2026

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Chapter 1

Introduction

The topic of this thesis centers around an open question about tensors. These are mathematical tools, used to describe relationships between multiple objects, allowing for some form of algebra to occur. In context of this work, the focus remains on how tensors can be built, where they lie, and how they can be broken down.

The state of Comon's conjecture (an open problem for the last 20 years) is not easily described. While mathematicians have chipped away at the number of cases where it remains valid, an answer in full generality over $\mathbb{R}$ or $\mathbb{C}$ remains unknown. The hope is that at least some of the work done in recent years can be highlighted as useful, paving the way towards a definitive counterexample or proof. This question merits study because of the powerful nature of tensors as descriptors of relationships. One of the more important problems that has come to light in recent years is the computational complexity of matrix multiplication ([18]), and tensors and their ranks help to quantify exactly how efficient the algorithms for matrix multiplication are. Tensors find their use in quantum mechanics (describing the entanglement of particles), general relativity (as tools to describe the curvature of space-time) [10], chemometrics, psychometrics (representing combinations of chemicals or traits that are interrelated)[28], and signal processing (the source selection problem)[18] — to name some more examples. In each of these cases, knowing how important a feature of the relationship symmetry is would play a vital role in analysing procedures and algorithms that take for granted that symmetry is the more "natural" state of affairs. This forms the guiding idea behind the ideas and techniques explored during this thesis.

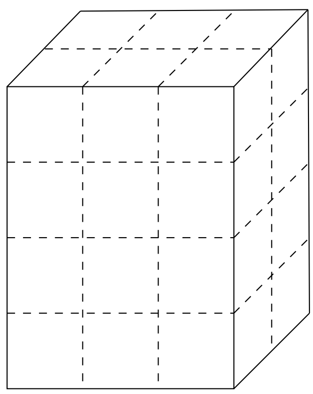

Figure 1: Example of a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$

Comon's Conjecture

At its heart, Comon's Conjecture centers around 2 central themes: rank and symmetry in tensors. Rank (see definition 2.2.3) describes how a tensor can be decomposed into simpler constituent tensors, and symmetry (see definition 2.3.2) describes how tensors behave under permutation actions.

The conjecture was first stated openly in July 2008 at ”Geometry and Representation Theory of Tensors for Computer Science, Statistics and Other Areas” - a workshop organised at the American Institute for Mathematics [19] in the group focusing on ”Multilinear Techniques in Data Analysis and Signal Processing”. The conjecture as stated in the report is as follows

Conjecture 1 (Comon)**.** *Do the rank and symmetric rank of T always coincide, where T is a real or complex tensor?*

The report also covered work done previously by Comon et al. [14] where they also showed that the conjecture held for situations when the symmetric rank was at least as large as the dimension, or when the order (see definition 2.1.1) of T is sufficiently large. Since 2008, the mathematical community has proved the conjecture true under various assumptions (see section 3.2).

In 2019, Shitov [22] proposed the construction of a counterexample living in $(\mathbb{C}^{800})^{\otimes 3}$ with symmetric rank at least 904, and a rank of at most 903. In 2024 however, an erratum published [15] showed a flaw in the construction, rendering the conjecture still open, at least for the complex case. In the intervening time, a paper published by Seigal and Wang [27] demonstrated the construction of a real, order 6, symmetric tensor with real rank 30913 and symmetric rank 30914, disproving the conjecture for the real case. This paper built on constructions described in Shitov [23], a preprint, whose ideas have been developed further in subsequent preprints [24] and [25].

The following chapters and sections highlight important terminology required to state and describe the setting of Comon's conjecture, highlight the work done to prove the conjecture in the past, and finally to describe the major constructions required to construct the real counterexample.

Chapter 2 This chapter covers preliminary terminology and definitions required to understand the history of attempts to prove or disprove Comon's conjecture. It also serves to show the different ways in which tensors can be viewed.

Chapter 3 This chapter goes over the literature that currently exists since the statement of the problem in 2008. The papers covered highlight the valid attempts at proving Comon's conjecture over $\mathbb{R}$ and $\mathbb{C}$

Chapter 4 This chapter focuses on understanding Shitov's methodology and approach to constructing a counterexample to Comon's conjecture. The culmination of this chapter is an examination of Wang and Seigal (2023) [27], a published counterexample to the conjecture.

Chapter 5 This chapter highlights the gaps present in the literature so far, illustrates problems with the current approach to constructing counterexamples, and seeks to highlight potential future avenues to solving this problem once and for all.

The personal contributions in this thesis are in the compilation of existing literature regarding Comon's conjecture, establishing lemmas required for the constructions made by Shitov, and generalising (where possible) the arguments made by Wang and Seigal.

Chapter 2

Preliminaries Regarding Tensors

2.1 Tensors in an Algebraic Setting

The most abstract way of thinking about tensors is when describing them as elements of the tensor product of modules. Consider A, B and P as being R-modules, where R is a ring. Then any mapping $t : A \times B \to P$ that is bi-linear in nature is described by acting on elements in this manner: $t(\lambda a, b) = t(a, \lambda b) = \lambda t(a, b)$, where $a \in A, b \in B, \lambda \in R$ and $t(a, b) \in P$.

$$
\begin{tikzcd}
A \times B \arrow[rr, "t \text{ bilinear}=\tilde{t}\circ\varphi"] \arrow[dr, "\varphi"'] & & P \\
& A \otimes_R B \arrow[ur, "\exists!\tilde{t}"'] &
\end{tikzcd}
$$

Figure 2: Illustrating the tensor product of R-modules

Such a bilinear (or even multilinear mapping) between R-modules can be represented by an element lying in their tensor product, such that the following diagram of modules commutes [26] (illustrated specifically for the case mentioned above in Figure fig. 2)

While such a description is useful when showing about how any general elements lying in spaces can be combined, this description lacks some of the necessary details required for actually computing what a tensor *is*, or even how it can be represented.

A more useful approach to tensors is to begin from linear algebra. As matrices represent linear transformations between vector spaces V and W, so d-way arrays can be used to represent multilinear relationships between vector spaces $V_1, V_2 \dots V_d$. Diving deeper into this analogy, matrices do behave as either bilinear maps from the product $V \times W \to \mathbb{F}$, or as linear maps $W \to V^*$. This hybrid nature of matrices is in fact a reflection of the fact that matrices are order 2 tensors. Since computing elements of the matrix A_{ij} depends on a choice of bases for both vector spaces V and W, an analogous choice of bases must be made to describe the order d tensor $T \in V_1 \otimes V_2 \otimes \cdots \otimes V_d$.

For the purposes of this thesis, unless specified, each vector space will be considered over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, with its standard ordered basis $\{e_i\}_{i=1}^n$ where $e_i(j) = \delta_{i,j}$.

Accordingly, each vector space V_i has a dimension n_i.

Definition 2.1.1 (Order of a Tensor)**.** A tensor T is said to be a d-order tensor, if $T \in V_1 \otimes \cdots \otimes V_d$.

Let $T := v_1 \otimes \cdots \otimes v_d$ be an order-d tensor, with each component $v_i \in V_i$ a vector with components $v_i(j)$ according to the standard ordered basis of the corresponding vector space. Then the elements of the d-way array that corresponds to T can be given in the following manner

$$T(k_1, k_2, \dots, k_d) = \prod_{i=1}^{d} v_i(k_i),$$

where $0 \leq k_i \leq n_i$— the dimension of each corresponding vector space V_i.

T as defined above can be seen in the following manner

1. as a multilinear mapping between $V_1^* \times V_2^* \times \cdots \times V_d^* \to \mathbb{F}$ (extending the analogue of looking at matrices as bilinear forms)

$$(f_1, f_2 \dots f_d) \mapsto \prod_{i=1}^{d} f_i(v_i)$$

2. as a linear mapping from $V_i^* \to V_1 \otimes \cdots \otimes V_{i-1} \otimes V_{i+1} \otimes \cdots V_d =: V_{\hat{i}}$ (moving from a space of order-d tensors to order-$d-1$)

$$f_j \mapsto f_j(v_i)(v_1 \otimes \cdots \otimes v_{i-1} \otimes v_{i+1} \otimes \cdots \otimes v_d)$$

This viewpoint can be seen in section 4.1.3, slicing the tensor along the ith index

3. as the dual of this previously defined linear map, now going from $V_{\hat{i}}^* \to V_i$ (moving now from order-d to a vector collinear to v_i)

$$(f_j)_{j \neq i} \mapsto \left(\prod_{j \neq i} f_j(v_j) \right)$$

Each of these above viewpoints also corresponds to an appropriate transformation of the representative d-way array chosen, either to a scalar, $d-1$-way array or a vector respectively [18]. In fact scalars and vectors now can also be seen as tensors of orders 0 and 1 respectively.

2.2 Rank

While the previous two sections covered the nature of tensors, it is necessary to figure out how tensors can be constructed, since the Segre embedding example shows that there are tensors that cannot be constructed as simply the tensor product of vectors — as an analogy, recall not all vectors in a vector space can be written as only scaled basis vectors.

Definition 2.2.1 (Elementary Tensor)**.** Consider a tensor $T := v_1 \otimes \cdots \otimes v_d \in V_1 \otimes \cdots \otimes V_d$, an order d tensor. Since T can be written explicitly as the tensor product of d vectors, it is known as an elementary tensor. Equivalent names for such tensors present in literature are *rank one* and *decomposable*.

Using these elementary tensors as building blocks now provides a way to describe the construction of more general tensors.

Definition 2.2.2 (Tensor Decomposition)**.** Consider a general tensor $T \in V_1 \otimes \cdots \otimes V_d$. A decomposition of T consists of writing T as a sum of elementary tensors

$$T = \sum_{i=1}^{s} v_1^i \otimes \cdots \otimes v_d^i$$

now with each of the $v_j^i \in V_i$. [28]

For any given tensor T now, there can exist multiple decompositions, not just depending on the choice of basis vectors for each V_i. A further natural question to ask is if there is a shortest decomposition, and if such a decomposition is unique.

Definition 2.2.3 (Tensor Rank)**.** The rank of a tensor T is the minimal length of any possible decomposition of T.

Tensor rank always exists, because tensors can always be written in terms of elementary tensors. Computing the rank of a given tensor is not an easy problem, but it has given rise to multiple approaches in the literature that rely on both the algebraic properties of tensors and associated transformations, and the geometric properties of the varieties on which these tensors lie. Tensor rank also remains invariant of decomposition and is one of the two key ingredients required to state Comon's Conjecture.

A further question that can be asked is whether "perturbing" a tensor changes its rank. This leads to the notion of describing rank when it is seen as the limit of a sequence.

Definition 2.2.4 (Border Rank)**.** A tensor T has border rank r if it can be expressed as a limit of a sequence of tensors of rank r, but not for any sequence of tensors with rank $s < r$. Conventional notation is to denote border rank with $\underline{\text{rank}}\,(T) = r$

The notion of border rank comes in use when working with secant varieties, as a tool to help compute rank. Tensors can have differing ranks and border ranks

Example 1 ([18])**.** Consider $T := a_1 \otimes b_1 \otimes c_1 + a_1 \otimes b_1 \otimes c_2 + a_1 \otimes b_2 \otimes c_1 + a_2 \otimes b_1 \otimes c_2$ Then the rank of this tensor is 3, but one can write T as the limit of tensors of rank 2 in the following manner

$$T(s) = \frac{1}{s}\left((s-1)a_1 \otimes b_1 \otimes c_1 + (a_1 + sa_2) \otimes (b_1 + sb_2) \otimes (c_1 + sc_2)\right)$$

2.3 Symmetric Tensors

The other half of Comon's Conjecture revolves around tensors that exhibit symmetric properties. These become valuable in applications simply because the symmetry means that they require "less" information to be constructed. What makes a tensor symmetric is the next logical question to tackle, and necessitates a definition of how permutations act on a given tensor.

Given $\sigma \in S_d$ (the symmetry group of $\{1, 2, \ldots, d\}$), where σ is a permutation of d distinct elements, one can define the permutation action on a tensor as follows

Definition 2.3.1 (Permutation of a Tensor)**.** Let $T := v_1 \otimes \cdots \otimes v_d \in V_1 \otimes \cdots \otimes V_d$, where $V_i = V_j$. Then $\sigma \in S_d$ acts on T in the following manner

$$\sigma \cdot T = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(d)}$$

Note here that we do require that all the vector spaces are the same, to keep the mapping well defined. This also allows for the use of notation $T \in V^{\otimes d}$.

The definition of a symmetric tensor follows almost immediately

Definition 2.3.2 (Symmetric Tensor)**.** Consider a tensor $T \in V^{\otimes d}$. T is said to be symmetric if the following equality holds

$$T = \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot T$$

The requirement of $T = \sigma \cdot T$ for all $\sigma \in S_d$ follows as a consequence of this definition. [14].

Further, since every tensor can be expressed as a decomposition (see definition 2.2.2), it follows that symmetric tensors can also be decomposed into a sum of elementary *symmetric tensors*, where now each is of the form $v^{\otimes d}$ by symmetry requirements.

Example 2. Consider the tensor $T = (1,0,0) \otimes (1,0,0) \otimes (1,0,0) + (0,0,1) \otimes (0,0,1) \otimes (0,0,1) + (0,1,0) \otimes (0,1,0) \otimes (0,1,0)$. This is definitely a symmetric decomposition, for T a symmetric tensor.

It is clear that not every tensor can be seen as symmetric, and therefore the subset of all order d tensors that are symmetric is denoted $S^d(\mathbb{F}^n) \subseteq \mathbb{F}^n$ — notation usually reserved for homogenous polynomials in n variables of fixed degree d. In fact one can show that these spaces are isomorphic to each other. This equivalence is a powerful tool that allows for an interchange of polynomial and tensor notation (when appropriate).

Symmetric Tensors and Homogenous Polynomials of Fixed Degree

A polynomial $f \in \mathbb{F}[x_1, \ldots, x_n]_d$ is of the form

$$f(x_1, \ldots x_n) = \sum_{a_i \geq 0, \sum_i^n a_i = d} \lambda_{(a_1, \ldots, a_n)} \prod_{i=1}^{n} x_i^{a_i},$$

where all the coefficients $\lambda_{(a_1,\ldots,a_n)} \in \mathbb{F}$. Here the following notation is commonly used in the literature — $\mathbf{x} = (x_1, x_2, \ldots x_n)$ and $\alpha = (a_1, a_2, \ldots a_n) \in (\mathbb{N} \cup \{0\})^n$. Then $\mathbf{x}^\alpha = \prod_{i=1}^n x_i^{a_i}$. Further $|\alpha| = \sum_i^n a_i$, so the homogenous polynomial as stated above can be rewritten as

$$f(\mathbf{x}) = \sum_{|\alpha|=d} \lambda_\alpha \mathbf{x}^\alpha.$$

The following proposition is standard in tensor decomposition literature, but is stated again for convenience.

Proposition 2.3.1. *For fixed values of $d, n > 0$, there is a mapping*

$$\Phi_d : \mathbb{F}[x_1, \ldots, x_n]_d \to S^d(\mathbb{F}^n),$$

an isomorphism of vector spaces over $\mathbb{F}$.

Proof. Define the mapping first for all degree d monomials $\mathbf{x}^\alpha$ as

$$\mathbf{x}^\alpha := \prod_{i=1}^n x_i^{a_i} \mapsto \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot (\bigotimes_{i=1}^n e_i^{a_i}).$$

Thus each monomial is mapped to a symmetric tensor (by construction). Here each monomial can be seen as a basis vector in the vector space $\mathbb{F}[x_1, \ldots, x_n]_d$.

To make sure this mapping is well defined, it is necessary to also emphasise that Φ must be a linear mapping (i.e. it must preserve addition and scalar multiplication in $\mathbb{F}[x_1 \ldots x_n]_d$). This ensures that every degree d homogenous polynomial has only one image, and establishes Φ as a linear map.

With this definition, Φ is an homomorphism of vector spaces because for $\lambda \in \mathbb{F}$

$$\Phi(\lambda \mathbf{x}^\alpha + \mathbf{x}^\beta) = \lambda \left(\frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot (\bigotimes_{i=1}^n e_i^{a_i}) \right) + \left(\frac{1}{d!} \sum_{\sigma \in S_d} \sigma \cdot (\bigotimes_{i=1}^n e_i^{b_i}) \right) = \lambda \Phi(\mathbf{x}^\alpha) + \Phi(\mathbf{x}^\beta).$$

Further, every elementary symmetric tensor $v^{\otimes d} = (\sum_{i=1}^n v_i e_i)^{\otimes d}$ can be seen as the image of the d-th power of a linear form — $(\sum_{i=1}^n v_i x_i)^d$. Since every symmetric tensor has a symmetric tensor decomposition, Φ must also be surjective.

Finally, consider two distinct monomials $\mathbf{x}^\alpha, \mathbf{x}^\beta \in \mathbb{F}[x_1, \ldots x_n]_d$ with $a_j \neq b_j$ for at least one $0 \leq j \leq n$. Their images then are also distinct under Φ because every term in the sum of the image is distinct. □

Thus there exists an isomorphism between these two vector spaces. In fact Φ_d can be seen as a component of a larger isomorphism between graded vector spaces [14].

Chapter 3

Literature Review

Since the stating of the problem in 2008, there has been a substantial amount of work done to arrive at a conclusive final resolution. This body of work relies on the geometric construction of secant varieties and how they relate to the question of computing rank and symmetric rank.

3.1 Secant Varieties

The notion of studying secants to a given geometric object is not novel, but doing so sheds more light on the description of tensors as lying in projective space. Considering a curve in a vector space, the secant line joining any two distinct points is described exactly by all the linear combinations of the points. This analogy is the inspiration for the following definition.

Definition 3.1.1 (r-th Secant Variety to X)**.** Let $X \subset \mathbb{P}V$ be a projective variety. Then the r-th secant variety to X, denoted $\sigma_r(X)$ is defined as

$$\sigma_r(X) := \overline{\bigcup_{x_1,\ldots,x_r \in X} \mathbb{P}_{x_1,\ldots,x_r}}$$

Here $\mathbb{P}_{x_1,\ldots,x_r}$ (for $x_1, \ldots, x_r$ lying in general position) is the projectivisation of the r-dimensional hyperplane defined by these r points. The bar denotes the Zariski closure in $\mathbb{P}V$, covering the cases where the r points chosen may not lie in general position, particularly also when all the r chosen points coincide leading to the corresponding tangent hyperplane.

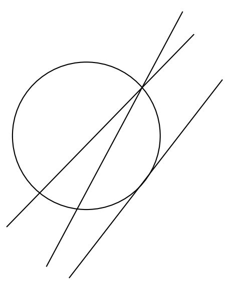

Figure 3: Secant lines through a circle (and a limiting tangent line)

Observation. It is clear from the definition above that successive secant varieties are nested, with $\sigma_0(X) = X$, $\sigma_1(X)$ describing all the tangent lines to X, and so on, such that $\sigma_{r-1}(X) \subseteq \sigma_r(X)$. Such a sequence of nested geometric objects must stabilise, for reasons of dimension.

Remark. While this geometric object is easy to describe with the notation above, computing the ideals for such a variety, or even the equations required to cut such a variety out form about half of the body of work surrounding Comon's conjecture. The following papers [8, 9, 4] develop alternative geometric tools to secant varieties that are easier to compute.

Since elementary tensors can be seen as lying on either the Segre or Veronese embedding, the r-th secant varieties to each of these describe all the tensors of *border rank* at most r. Note here that it is border rank that because of the closure in the definition of the secant variety. To describe all tensors of *rank* at most r, it is enough to look at the *interior* points of the r-th secant variety, this interior again being under the Zariski topology on $\mathbb{P}V$.

3.2 Overview of Symmetric Tensor Decomposition

As mentioned in the introduction Comon et al. [14] showed symmetric rank and rank coincided for tensors of ”large enough” order — a notion that became concrete in later literature. The authors also point out as a result of the Veronese embedding (see **??**), the rank of a symmetric tensor $T \in S^d(\mathbb{C}^n)$ cannot be more than $\binom{n+d-1}{d}$.

Since $S^d(\mathbb{C}^n) \subseteq (\mathbb{C}^n)^{\otimes d}$, they pointed out that for all $T \in S^d(\mathbb{C}^n)$, rank $(T) \leq$ srank (T). Their major result was that the conjecture holds ”generically” (i.e almost everywhere) whenever srank $(T) \leq n$, the dimension of the underlying vector space. In fact for every $T \in S^d(\mathbb{C}^n)$, they showed that Comon's conjecture holds if srank $(T) = 1, 2$.

Brachat et al. [6], in 2010 proposed a new algorithm for symmetric tensor decomposition beginning with the relationship between symmetric tensor decomposition and the secant varieties of a Veronese variety (see Section section 3.1). Their resulting algorithm required a reformulation of the problem in a dual space, and Hankel operators.[INCLUDE THIS?]

A convergent approach to Comon's conjecture arises from a classical problem in number theory — the Waring decomposition of a number. This problem then gets restated into the Waring problem for polynomials as follows: *What is the the minimum number of linear polynomials required such that a linear combination of them to the dth power is equal to a homogenous degree d polynomial?*

A solution to the polynomial Waring problem definitely solves the first issue of computing symmetric tensor decompositions (see section 2.3). This problem was solved by Alexander and Hirschowitz in 1995, where they showed that $\sigma_r(\nu_d(\mathbb{P}\mathbb{F}^{n+1}))$ - the r-th secant variety to the d-th Veronese embedding of $\mathbb{F}^{n+1}$ - is of expected dimension with 5 exceptional cases. Brambilla and Ottaviani [7] in 2008 gave a self-contained proof of the theorem, with some simplifications, particularly in the case of order 3 symmetric tensors.

Landsberg [18], in his textbook describes the conjecture in terms of (border) rank preserving pairs of projective varieties.

Definition 3.2.1 ((Border) Rank Preserving Pair)**.** $X \subseteq \mathbb{P}V$, a variety, and $L \subseteq \mathbb{P}V$ a linear subspace when taken together as (X, L) are a *(border) rank preserving pair* if $\langle (X \cap L)_{red} \rangle = L$ and rank $(X|_L) =$ rank $(X|_{(X \cap L)_{red}})$. (A similar condition holds for border rank, but this can now be stated in terms of secant varieties as $\sigma_r(X) \cap L = \sigma_r((X \cap L)_{red})$. Here $(X \cap L)_{red}$ refers to the intersection without any points of non-unit multiplicity. and rank is now seen as a function from $\langle X \rangle \to \mathbb{N}$

Remark. This translates the statement of Comons conjecture from algebra to a statement about (secant) varieties lying in some projective space. In this restatement, $\Sigma(\mathbb{P}\mathbb{F}^n \times \cdots \times \mathbb{P}\mathbb{F}^n)$ (the d-fold Segre embedding) and $\mathbb{P}(S^d(\mathbb{F}^n))$ (all the lines passing through the d-th Veronese) are required to be a rank-preserving pair.

Ballico and Bernardi [2] in 2013 showed that that tensors lying on the tangent variety to a Veronese variety (see **??**) must always have rank $=$ srank . Their proof highlighted explicit algorithms for the computation of the rank of a tensor belonging to a given secant variety. Their computation involves describing how a point on the tangent variety can be expressed by lying on the linear span of a degree 2 zero-dimensional scheme.

While secant varieties are convenient geometric tools to use, Buczynska and Buczynski[8] demonstrate how the commonly used practice of using the $(r+1) \times (r+1)$ minors of the *catalecticant matrix* to compute the generating equations for the secant variety are not sufficient for higher order Veronese varieties (order $>> 2$). This paper resulted in the developement of *cactus varieties.*

Oeding and Ottaviani [20] in 2013 contributed new algorithms for computing solutions to the polynomial Waring decomposition problem - equivalent to the computation of secant varieties to a given Veronese variety. They did so by further developing work done to define the eigenvectors of a tensor. Using these eigenvectors and Kozul flattenings allowed for an algorithm that could compute the Waring decomposition while also remaining valid for previously covered cases under the catalecticant matrix methodology.

Computing decompositions covers one half of the questions about symmetric rank. The other half concerning the uniqueness of a decomposition requires the definition of when a tensor is *identifiable*.

Definition 3.2.2 (Identifiability)**.** A tensor T is said to be identifiable if it admits only one rank decomposition (modulo scaling by d-th roots of unity and permutations of summands)

Observation. This definition is equivalent to asking that an identifiable tensor lie on only one secant variety (again modulo permutations of points on the Segre embedding, and scaling factors)

Identifiability is a useful property when working with symmetric tensors because no further computation is required once a tensor decomposition is found. Particularly for the case of Comon's conjecture, the distinction between rank and symmetric rank (if such a tensor exists) implies that the tensor in question is *not* identifiable.

Generic identifiability is the idea that almost every point (i.e. a point in a dense open subset), is identifiable. Specific identifiability poses the opposite question - given a specific tensor T, does it admit a unique decomposition? Chiantini, Ottaviani and Vannieuwenhoven [13, 12] in 2 subsequent papers tackled both of these questions. Their earlier paper showed that a general symmetric tensor in $S^d(\mathbb{C}^{n+1})$ of sub-generic rank r was identifiable, provided $r \leq \binom{n+d}{d}/(n+1)$ (with exceptions again being covered by the Alexander-Hirschowitz theorem). Their second paper focused on showing what criteria could be called "effective" to determine if a given tensor was specifically identifiable. Criteria were designated effective if their conditions were satisfied by generic points (i.e. they were satisfied in an open, dense subset). One of the conclusions this paper reacheed was that Kruskal's criterion fell in line with this definition of "effectiveness".

Friedland [16] in 2016 focused not on the computation of the symmetric rank, but on the properties of symmetric tensors of a given symmetric rank. In particular the paper focuseed on showing that when a symmetric tensor can be unfolded into a matrix (see section 4.1.4), then Comon's Conjecture held if the rank of the tensor when computed was either the rank of the matrix or the rank of the matrix $+1$.

Zhang et al. [28] conclusively showed that Comon's conjecture held for all tensors where the rank of the tensor under consideration is not more than its order.

Ballico et al [3] at the end of 2018, refocused on the question of identifiability, and worked on refining Kruskals criterion (sharp and algebraic in nature) into a geometric criterion that can be verified more easily. They concluded that Comon's conjecture holds for general tensors whose symmetric rank r is bounded above strictly by $\binom{n+e}{e}$ for a specific choice of e. This choice depended upon splitting the Segre product into two disjoint components and then imposing conditions on a set of points on the Segre that exhibited distinct behavious when projected down to the first component versus the second . The authors of this paper clearly point out that even though they have increased the scope of validity of Comon's conjecture, Shitov's proposed counterexample [22] lies outside this scope and thus might have still been a plausible counterexample.

Bernardi et al. [5] in 2018 comprehensively covered the literature and algorithms required to understand how secant varieties become useful in establishing questions about (generic) rank for tensors, with an emphasis on symmetric tensors and the mapping to the Veronese variety.

Casarotti et al. [11] proposed to improve the bounds for Comon's conjecture when working with flattenings, provided that equations for the secant varieties associated to the Veronese variety could be generated. Their result was that $h < \binom{n+\lfloor \frac{d}{2} \rfloor}{n}$, for a tensor having rank h implies that srank $= h$, for cases where such generating determinantal equations were known.

Finally in [21] Seigal demonstrated that for all "cubic surfaces", Comon's conjecture does hold. (even up to border rank.) Here cubic surfaces were used to refer to the fact that under the identification with homogenous polynomials, order 3 symmetric tensors correspond to cubic polynomials, and thus their zero sets could be called cubic surfaces. She also used flattenings of tensors (see section 4.1.4), dividing her proof into cases depending on the number of singularities on the cubic surfaces in question.

3.3 Overview of Valid Cases of the Conjecture

The current overview of the cases where Comon's Conjecture holds so far is as follows:

- For $T = \sum_{i=1}^{s} y_i^{\otimes d}$ and d "large enough" [14]. Here the large enough argument in the paper hinges on a classical result from algebraic geometry, stating that d points in $\mathbb{C}^n$ form the solution set of of polynomial equations of degree $\leq d$
- For $T \in S^d(\mathbb{C}^n)$ and rank$_S(T) = 1, 2$ [14]
- For $T \in S^d V$, and rank$(T) \leq \frac{dn-d+1}{2}$ where V is a real or complex vector space with dimension n [19]
- For $T \in S^d(\mathbb{C}^n)$ for $(d, n) \in \{(3, 2), (4, 2), (3, 3)\}$ [16]
- In the case $d \geq 3, |\mathbb{F}| \geq 3, T \in S^d(\mathbb{F}^n)$ and rank$(T) \leq$ rank$U(T) + 1$ (Here U denotes the flattening section 4.1.4 of T in any direction, since T is symmetric all directions are equivalent) [16]
- When $T \in S^3(\mathbb{C}^n)$ and rank$(T) \leq 5$ or rank$_S(T) \leq 6$ [16]
- For $T \in S^d(\mathbb{F}^n)$ where rank$(T) \leq d$ where $\mathbb{F} = \mathbb{R}, \mathbb{C}$[28]
- If T is a tensor belonging to a tangential variety to a Veronese variety [2]
- If $T \in \mathbb{C}^2 \otimes \mathbb{C}^n \otimes \mathbb{C}^n$ [9]
- For T being a Coppersmith-Winograd tensor [17]
- Casarotti [11] sets up two propositions that suggest whenever determinantlal equations for secant varieties can be written down, Comon's conjecture should be *verifiable*
- Whenever a tensor T is cubic surface (real or complex) [21]

Chapter 4

Constructions required for a Counterexample

4.1 History

In spite of all of these preliminary results that indicate positive answers towards Comon's Conjecture, Yaroslav Shitov seemed to arrive at a method of constructing a counterexample, as early as 2019 [22]. An apparent error found in this paper leads to an inalidation of this result in 2024, but the period of 5 years in the interim led to multiple other preprints from Shitov, starting with a newer, alternate methodology for the construction of counterexamples. Therefore, while there is now no material carrying the stamp of peer-review and Shitov's name, his contributions and subsequent papers seem to indicate a viable approach to disproving Comon's conjecture, at least over $\mathbb{R}$ if not $\mathbb{C}$.

4.1.1 Shitov's First Paper in 2019

The 2018 paper begins with taking the ground field as $\mathbb{C}$, and fixing the notation to be used. The main objects of interest are order 3 tensors, and the introductory section of the paper highlights the behavious of such tensors under elementary transformations with respect to their constituent order 2 ”slices” (see section 4.1.3). The construction of the counterexample begins in earnest with the description of a 100×100 matrix, partitioned into 20 submatrices. Each submatrix then is mapped to a corresponding 100×100 matrix, now with entries in the submatrix seen as 1 and all other entries as 0. Taking the $1-$ and $2-$ supports of each of these, and pre- and post-multiplying them with a column-shift operator with parameter a leads to another set of matrices of similar size (still 100×100).

It is at this point that the paper makes a construction that seems strange. A set is constructed consisting of matrices of distinct formats (viz. 200×200 and 400×400). While there is no issue in defining such a set, the paper then constructs ”the $\mathbb{C}$-linear span” of the elements of this set. Addition of quantities that do not live in the same dimension (at least from this author's prior knowledge), without a specified or implied embedding map presents the first major problem when following this paper.

The preceding instructions illustrate the procedure followed by the paper, but later in the paper a tensor decomposition is claimed to exist of the following form (no context is necessary for the symbols used, it is enough to know they indicate tensors of appropriate formats being combined)

$$\mathcal{S} - \Phi = \Xi_1 + \Xi_2 + \Xi_3 + \sum_{b \in \mathcal{B}'} (\Xi_b^1 + \Xi_b^2 + \Xi_b^3)$$

This statement is the subject of the erratum published in 2024 [15] indicating such a statement was true only in cases where $\Xi_b = 0$, but equality could not be said to hold in general otherwise without proof. This crucial observation led to the invalidation of this entire construction.

While certain choices were made during this process that led to unanswered questions, the overall tools for establishing a counterexample were first established in this paper. These include the two minor operations - slicing tensors (section 4.1.3) and flattening tensors (section 4.1.4) and the three major operations - adjoining slices to tensors (section 4.2.1), generating families modulo slices (section 4.2.3) and cloning (section 4.2.4).

4.1.2 Shitov's Subsequent Preprints

In the following years, Shitov put out a number of preprints [23, 24, 25] each either tackling Comon's conjecture in a restricted setting, or developing tools to construct more general counterexamples, that might also be used to construct a counterexample to Strassen's conjecture (another rank conjecture concerning the additivity of rank.) The preprint from 2020 [23] tackles the conjecture over $\mathbb{R}$ and it is this that forms the basis for the paper from Wang and Seigal[27].

The constructions made in each preprint follow the same pattern, indicating a common toolset and structure of approach. Each approach sees an order d tensor as composed of lower order $d-1$ tensors, which allows for tensors to be transformed per "slice" along a particular direction. Viewing tensors in this manner opens a way to describe tensors modulo these slices — a useful operation, seen in both number theory and algebraic geometry as a tool to study problems in a more localised setting. Such a family, when generated, gives an idea of the general properties of the orginal tensor (how many linearly independent slices in which direction etc.) Flattenings assist with the computation of rank for these slices (computing the rank of matrices is a known and easier problem to solve than doing so for higher order tensors) and lend themselves to descrbing rank information that might change in the relationship between a tensor and its slices.

These tools allow for an analysis of any particular tensor, but the "construction" aspect of this approach becomes evident with the adjoining of slices to a tensor. Such an operation changes the format of the original tensor with respect to certain "chosen" slices, and allows the rank information of the family generated modulo the slices to be preserved in a new tensor that now contains both the original and the slices that are required.

Cloning starts to play a role perturbing such an adjoined tensor (usually by the modification of one of the constituent slices) causes a violation of the properties that were supposed to be preserved. Cloning is a way to maintain rank information by simply making a higher format copy, and allows for more "freedom" when modifying slices.

While each of these preprints starts with this general structure, and [25] proposes the construction of a general counterexample over any $\mathbb{F}$ with char$(\mathbb{F}) \neq 3$, an intermediate step involved in all of these is the construction of an "emulating family". Such a family, if it exists, allows for the rest of the constructions to proceed. However, proving the existence of such a family is not trivial. The latest preprint does attempt to do so, highlighting a similar construction from [24], but tracing the construction back leads to the definition of a function dependent on an arbitrary constant - in the same way as the paper [22] relied on the existence of an arbitrarily partitioned 100×100 matrix. Such choices, while perhaps valid, did not appear to be explained at any point in their corresponding documents, raising some skepticism and prompting further investigation into better documented methods that could explain their existence, or proofs that could deny their existence.

4.1.3 Slicing a Tensor

Viewing the columns or rows of a matrix as linearly dependent or independent and how they affect the overall linear transformation described by the matrix is the exact low-order analogue for describing tensors as composed of lower-order slices. Conventionally the columns of a matrix would be the 1 slices.

Definition 4.1.1 (Slices of a Tensor)**.** Let $T \in \mathbb{F}^{n_1} \otimes \cdots \otimes \mathbb{F}^{n_d}$ be an order d tensor. Then the j-th i slices of T are given by

$$T^i_j := f_j(v_i)(v_1 \otimes \cdots v_{i-1} \otimes v_{i+1} \cdots \otimes v_d)$$

$f_j(v_i)$ denotes the jth component of v_i and $1 \leq j \leq n_i$. For each *mode* $1 \leq i \leq d$, there are n_i order $d-1$ *slices* that make up T.
This definition is equivalent to selecting the j-th i-slice of the d-way array that represents T

Observation. Through the rest of this thesis, the superscript accompanying a tensor variable indicates the mode chosen along which any operation must occur.

Example 3. Consider a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$, where $T := (1,2,3,4) \otimes (1,2,3) \otimes (1,2)$. Then slices along different modes are computed as follows (visualised in fig. 4)

- $T^1_3 = 3((1,2,3) \otimes (1,2))$
- $T^2_2 = 2((1,2,3,4) \otimes (1,2))$
- $T^3_2 = 2((1,2,3,4) \otimes (1,2,3))$

Observation. Slicing in the above definition is described for only an elementary order d tensor, but it is a linear operation (addition and scalar multiplication don't change it.)

Lemma 4.1.1. *The operation defined above for taking the j-th i-slice of an order d tensor $S^i_j : V := \mathbb{F}^{n_1} \otimes \cdots \otimes \mathbb{F}^{n_d} \to V_{\hat{i}}$ defined as $T \mapsto T^i_j$ is a linear map between vector spaces.*

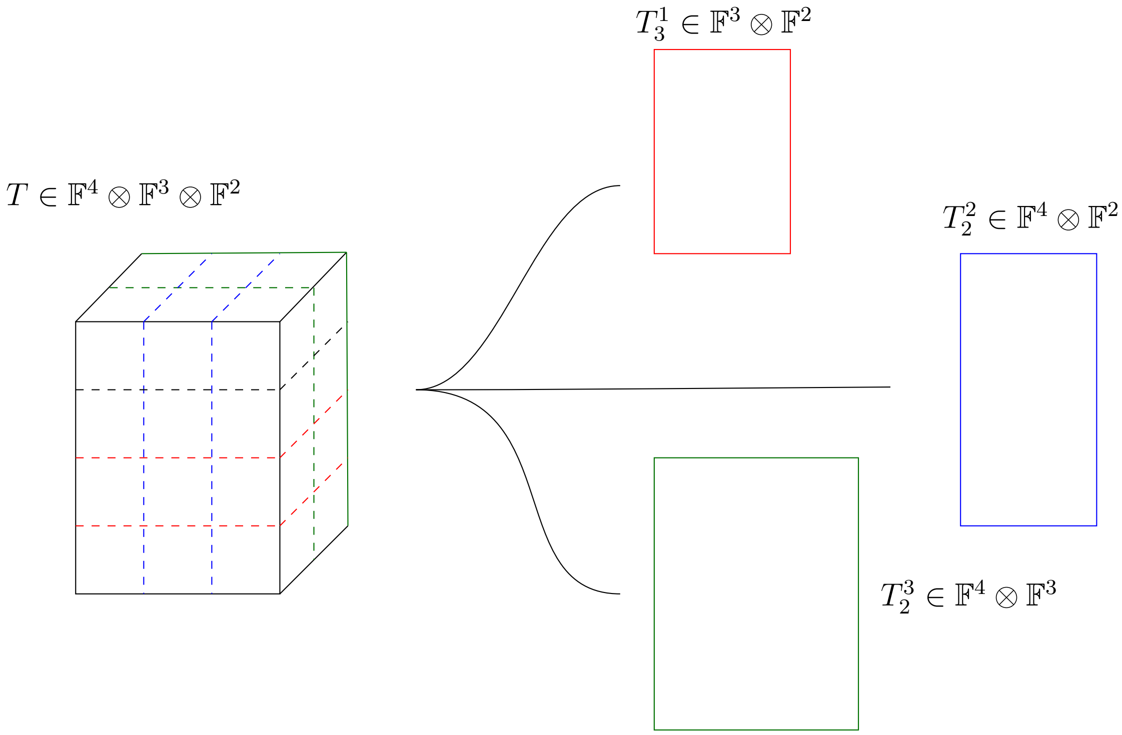

Figure 4: Slicing a tensor T along different modes

Proof. Let $T := \otimes_{i=1}^d v_i \in V$ be an elementary tensor. Then $S_j^i(T)$ exists and lies in the appropriate $V_{\hat{i}}$. Further for $T_1 = T_2 \in V$ one has $S_j^i(T_1) = S_j^i(T_2)$ by applying the definition provided above. Consider $S_j^i(T_1+T_2)$. Then the j-th i-slices of the d-way array T_1+T_2 are exactly given by $f_j(v_i)(v_{\hat{i}})+f_j(w_i)(w_{\hat{i}})$ where $v_i, w_i \in \mathbb{F}^{n_i}$ and $v_{\hat{i}}, w_{\hat{i}} \in V_{\hat{i}}$. Thus S_j^i as an operation preserves addition. Likewise $S_j^i(\lambda T) = \lambda(S_j^i(T)) = \lambda T_j^i$, preserving scalar multiplication. Therefore S_j^i is a linear operation $V \to V_{\hat{i}}$. □

Lemma 4.1.2. *Given a tensor T such that rank $(T) = r$, then there is at least one mode $1 \leq i \leq d$ having r linearly independent slices T_j^i where $1 \leq j \leq n_i$.*

Proof. Suppose the claim is false i.e. there is no mode i such that the number of linearly independent slices T_j^i is r. Since we are only dealing with a finite order d, there is a mode i with the largest number of linearly independent slices T_j^i. Let this number be ρ.

$\rho < r$: Then since $\rho < r$, each of the slices $S_i^j(T)$ can be written as a linear combination of exactly ρ order $d-1$ slices. One can therefore rewrite the decomposition of T in terms of vectors in V_i and the linearly independent set of $S_j^i(T)$, a decomposition of length less than r, a contradiction to the definition of rank.

$\rho > r$ Then any decomposition of length r must miss out at least one of the linearly independent slices that make up T, a contradiction to the definition of a tensor decomposition.

□

Corollary 4.1.2.1. *Given a tensor T of rank r, the collections of slices $\{T_j^i\}$ along all modes $1 \leq i \leq d$ have r linearly independent elements i.e $\langle T_j^i \rangle_{j=1}^{n_i}$ is an r-dimensional subspace.*

When working with symmetric tensors, slicing along any mode results in a lower order tensor in the same format. In this case $S^i_j : (\mathbb{F}^n)^{\otimes^d} \to (\mathbb{F}^n)^{\otimes^{d-1}}$ for all modes i.

4.1.4 Flattening a Tensor

Flattening, while not as important in Shitov's constructions, is an important operation to verify rank information for tensors. Such an operation can generally be used as a sanity check when performing operations or constructions on tensors.

Definition 4.1.2 (Flattening a Tensor)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$, be an order d tensor, with $T = v_1 \otimes \cdots \otimes v_d$. Define $T^{(i)}$, the i-th flattening as follows

$$T^{(i)}((k_1, \dots k_{i-1}, k_{i+1} \dots k_d), k_i) := T(k_1, k_2, \dots, k_d)$$

As with slicing, this definition is equivalent to flattening the d-way array representing T into a matrix with the elements in a row indexed by $\{1, \dots, n_i\}$ and elements in a column indexed by $\times_{j \neq i}\{1, \dots n_j\}$

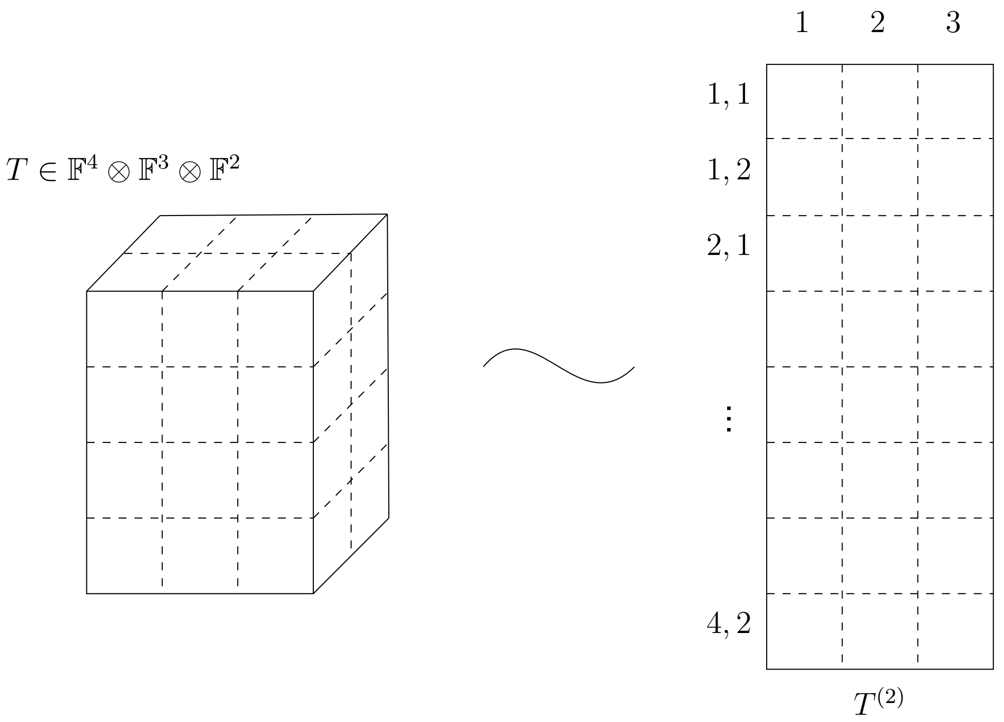

Figure 5: Flattening a tensor T

Example 4. Consider a tensor $T \in \mathbb{F}^4 \otimes \mathbb{F}^3 \otimes \mathbb{F}^2$, where $T := (1, 2, 3, 4) \otimes (1, 2, 3) \otimes (1, 2)$. Then flattening the tensor along the 2-mode can be computed as

$$T^{(2)} = ((1, 2, 3, 4) \otimes (1, 2))^{(1)} \otimes (1, 2, 3)$$

While this looks strange, the vector required to be tensored with $(1, 2, 3)$ is formed by flattening the matrix $(1, 2, 3, 4) \otimes (1, 2)$ along the 1 mode. This is equivalent to the vector $(1, 2, 3, 4, 2, 4, 6, 8)$. Figure 5 provides an illustration of this example, also highlighting how the column and row indices are generated.

Observation. Flattening T to $T^{(i)}$ is the operation by which (from section 2.1) T could be seen as a linear map between vector spaces instead. Here $T^{(i)} \in V_{\hat{i}} \otimes V_i$, seen as a matrix, representing a mapping from $V_i \to V_{\hat{i}}$ Each column $T_j^{(i)} := T^{(i)}(\dots|j)$ represents the "unravelling" of the j-th i slice of T, denoted $(T_j^i)^{\emptyset}$, into a vector.

Lemma 4.1.3. *The flattening operation* $F^{(i)} : V := \mathbb{F}^{n_1} \otimes \cdots \otimes \mathbb{F}^{n_d} \to W \otimes V_i, T \mapsto T^{(i)}$ *is a well defined linear mapping between vector spaces where* $W = \bigoplus_{j \neq i} \mathbb{F}^{n_j}$*.*

Proof. For any elementary tensor $T = v_1 \otimes \cdots \otimes v_d$, the flattening $F^{(i)}(T) = T^{(i)}$ exists in the appropriate vector space. Moreover if $T_1 = T_2$, then the corresponding images $F^{(i)}(T_1) = F^{(i)}(T_2)$, by the definition provided above, comparing each vector componentwise.

Consider $T_1 := \otimes_{i=1}^d v_i, T_2 := \otimes_{i=1}^d w_i$. Then $T_1 + T_2$ is a rank 2-tensor. $F^{(i)}(T_1 + T_2)$ is the same as $(v_{\hat{i}})^{\emptyset} \otimes v_i + (w_{\hat{i}})^{\emptyset} \otimes w_i = F^{(i)}(T_1) + F^{(i)}(T_2)$. Likewise $F^{(i)}(\lambda T_1) = \lambda(F^{(i)}(T_1)) = \lambda T_i^{(i)}$. Thus $F^{(i)}$ preserves both addition and scalar multiplication. □

Remark. In the literature surveyed, flattening can be done not just for one mode, but for a collection of modes. This gives rise to the empty set notation denoting the unravelling to a vector. Since this is less important for the following constructions, this more general definition of flattening is not covered.

Remark. For a tensor $T \in V_1 \otimes \cdots \otimes V_d$, $T^{\emptyset}$ denotes a vector lying in $V_1 \oplus \cdots \oplus V_d$. Here each component of the vector is indexed by the the set $\prod_{i=1}^d [n_i]$, and $T^{\emptyset}((k_1, \dots, k_d)) = T(k_1, \dots, k_d)$

When dealing with symmetric tensors, flattening along any mode is equivalent (since the tensor is symmetric), and so $T^{(i)} = T^{(j)}$ for all $i \neq j$. This resulting matrix is an element of $\mathbb{F}^{(d-1)n} \otimes \mathbb{F}^n$.

4.2 Major New Constructions

While the preceding constructions are relatively "minor" in the grand scheme of Shitov's proposed counterexample to Comon, the following three major operations on tensors are integral to the final construction proposed (and indeed implemented by Wang and Seigal [27]). The following sections deal with constructions involving the operations on a specific tensor parametrised by families of slices of lower order. It is convenient to represent such finite families of slices in terms of an appropriate order d tensor, and then reframe definitions accordingly in terms of this newly created order d tensor.

Definition 4.2.1 (Creating a Tensor from a Family of Slices)**.** Let $d > 0$ and $1 \leq i \leq d$. Take $\mathcal{M} := \{M_1, \dots, M_N\} \subset V_{\hat{i}}$ as a finite collection of order $d-1$ tensors. Since $\mathcal{M}$ is finite, it can be rewritten as a tensor of order d, lying in $V_1 \otimes \cdots \otimes V_{i-1} \otimes \mathbb{F}^N \otimes V_{i+1} \otimes \cdots \otimes V_d$ in the following manner where $\bar{\mathcal{M}} \in V_1 \otimes \cdots \otimes V_d$ is defined in terms of each of the i-slices M_j, where $1 \leq j \leq N$

$$\bar{\mathcal{M}}(k_1, \dots k_{i-1}, j, k_{i+1} \dots, k_d) := M_j(k_1, \dots k_{i-1}, k_{i+1} \dots k_d)$$

4.2.1 Background to Adjoining

Once the notion of a tensor of order d being composed of various order $d-1$ slices has been established, one natural operation that can be performed is the *adjoininig* of slices along the d modes of the tensor, provided that each slice is of appropriate format. To perform this operation, one needs two pieces of information — the tensor to be transformed, and the collection of slices (of appropriate format) that are to be adjoined. This operation seems to have been first formally defined by Shitov in his 2019 paper [22], and further developed into the language below in [23].

Definition 4.2.2 (Adjoining Slices to a Tensor along a specific mode)**.** Let $T \in V_1 \otimes \cdots \otimes V_d$ be an order d tensor, and $\mathcal{M} = \{M_1, \ldots, M_N\} \subset V_{\hat{i}}$ be a finite collection of order $d-1$ slices.

Define $\tau := Adj_i(T, M) \in V_1 \otimes \cdots V_{i-1} \otimes \left(V_i \oplus \mathbb{F}^N\right) \otimes V_{i+1} \cdots \otimes V_d$ in the following manner

$$\tau(k_1, \ldots, k_d) = \begin{cases} T(k_1, \ldots, k_d) & (k_i)_{i=1}^d \in \prod_{i=1}^d \{1, \ldots, n_i\} \\ \bar{\mathcal{M}}(k_1, \ldots, k_d) & (k_j)_{j \neq i} \in \prod_{j \neq i} \{1, \ldots, n_j\} \text{ and } n_i \leq k_i \leq n_i + N \end{cases}$$

Note that the above definition is still coordinate dependent. Define $\hat{V}_i := V_1 \otimes \cdots \otimes V_{i-1} \otimes \bigoplus_{\lambda \in \Lambda} \mathbb{F} \otimes V_{i+1} \otimes \cdots \otimes V_d$. Defining $\hat{V}_i$ in this manner allows for the following (re)definition of the adjoining operation, now in terms of a well defined function. Thus $Adj_i : V_1 \otimes \cdots \otimes V_d \times \hat{V}_i \to \hat{V}_i$ defined as taking $(T, M) \mapsto \tau$ as defined above, is a well defined function. It is now necessary to define an appropriate projection mapping, to allow the image of this map to be viewed as a tensor in finite dimensions along every mode (finite format?).

Definition 4.2.3 (Projecting to obtain a tensor of finite format)**.** Given an element $T' \in \hat{V}_i$, define the projection mapping

$$\pi_N^i : \hat{V}_i \to V_1 \otimes \cdots \otimes V_{i-1} \otimes \mathbb{F}^N \otimes V_{i+1} \otimes \cdots \otimes V_d$$

as restricting T' to an order d tensor of finite format in a natural way. In coordinates

$$\pi_N^i(T')(k_1, \ldots, k_d) = T'(k_1, \ldots, k_d), (k_i)_{i=1}^d \in [n_1] \times \cdots \times [n_{i-1}] \times N \times [n_{i+1}] \times \cdots [n_d]$$

Remark. The above two definitions allow for obtaining a tensor of finite format when adjoining a family of slices along only one mode. The next major step to be taken is adjoining slices along multiple directions. Doing so requires the definition of a ”padding” operation, that allows for lower order tensors to be adjusted into appropriate format, before being adjoined in the appropriate direction.

Definition 4.2.4 (Padding a lower order tensor)**.** Given $d > 0$ $1 \leq i \leq d$ as fixed. Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be an order d tensor, and $A \in W_1 \otimes \cdots \otimes W_{i-1} \otimes W_{i+i} \otimes \cdots \otimes W_d =: W_{\hat{i}}$ be an order $d-1$ tensor, such that $\dim(V_j) \geq \dim(W_j)$ for all $j \neq i$. Then

$$pad_i : W_{\hat{i}} \times V \to V_{\hat{i}}$$

as a function can be defined in coordinates as

$$pad_i(A, T)(k_1 \ldots k_{i-1}, k_{i+1} \ldots k_d) = \begin{cases} A(k_1 \ldots k_{i-1}, k_{i+1} \ldots k_d) & 1 \leq k_j \leq \dim(W_j) \forall j \neq i \\ 0 & \text{otherwise} \end{cases}$$

Combining all of these functions allows for adjoining families of $d-1$-order tensors along every mode of a given order d tensor according to the following procedure.

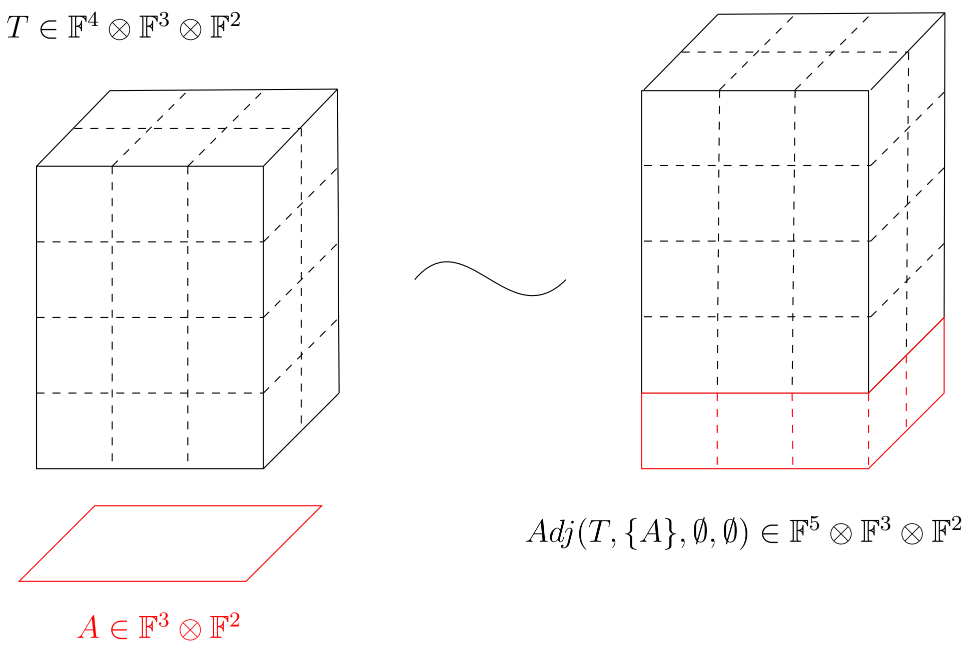

Figure 6: Adjoining a slice A to a tensor T

4.2.2 Adjoining Slices to a Tensor

Given an order d-tensor $T \in V$ and a collection of finite subsets of $d-1$ tensors $\mathcal{M}_i \subset V_{\hat{i}}$ for $1 \leq i \leq d$. Then $Adj(T, \mathcal{M}_1, \dots \mathcal{M}_d)$ can be constructed in the following manner

Step 1 $T_1 = \pi^1_{|\mathcal{M}_1|+n_1} \circ Adj_1(T, \bar{\mathcal{M}}_1)$ resulting in a new intermediate tensor with slices adjoined along the 1-mode, of finite format.

Step 2a Let $M_2 = \{pad_2(A, T_1) | A \in \mathcal{M}_2\}$, creating a set of appropriately padded slices, now to be adjoined along the 2-mode.

Step 2b $T_2 = \pi^2_{|M_2|+n_2} \circ Adj_2(T_1, \bar{M}_2)$

⋮

Step da $M_d = \{pad_d(A, T_{d-1}) | A \in \mathcal{M}_d\}$

Step db $T_d = \pi^d_{|M_d|+n_d} \circ Adj_d(T_{d-1}, \bar{M}_d)$

The tensor $T_d \in (V_1 \oplus \mathbb{F}^{|\mathcal{M}_1|}) \otimes \dots \otimes (V_d \oplus \mathbb{F}^{|\mathcal{M}_d|})$ is the required result of the process.

Observation. Figure 6 illustrates the above definition for the case when only one slice is being adjoined in one direction. According to the above definition τ can be seen as an order d tensor still, but now lying in $\left(V_1 \oplus \langle e^1_{n_1+1} \rangle\right) \otimes \dots \otimes V_d$ where $e^1_{n_1+1}$ indicates a "new" basis vector added onto the original vector space V_1.

The *symmetric adjoin* can be denoted $SAdj(T, \mathcal{M})$ where $\mathcal{M}$ is adjoined along every mode, following the same process as given in section 4.2.2.

The adjoining process as described creates a tensor of larger format, but the rank of such a tensor may be appropriately constrained depending on the slices being adjoined. It is this constraining property that Shitov and the authors of [27] use to construct an intermediate tensor with adjoined slices that has a differing rank and symmetric rank.

Remark. It may be trivial to note that while the order of the slices in $\mathcal{M}_i$ matters when constructing the final tensor, the rank information of the constructed tensor is independent of such changes in ordering. Further, the adjoining procedure itself can be modified, to follow any ordering of modes, as long as the slices required have been padded appropriately.

4.2.3 Generating Tensor Families modulo Slices

Given that a tensor admits many non-unique decompositions, one interesting way to analyse the structure of a tensor is in analogue to Gaussian elimination for matrices. The substitution method as described in [17, 1] has been a computational tool since the 1960's. Rather than arrive at one possible "lowest" optimal point, generating a family of tensors modulo a set of slices becomes equivalent to looking at all the possible ways a tensor could be transformed, based on transforming its constituent slices.

Defining such an operation that generates a linear span necessitates one additional description of padding a tensor into an appropriate format, but now in terms of seeing the lower order slice as embedded into a higher order tensor. In doing so, it is also necessary to have a candidate higher order tensor as a "template" that the slice can be made to resemble.

Definition 4.2.5 (Padding into higher order)**.** Let $d > 0$, and $1 \leq i \leq d$. Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be the candidate higher order tensor to be used as a template, and $A \in V_{\hat{i}}$ the order $d-1$ tensor to be padded into order d.

Then define $Pad_j^i : V_{\hat{i}} \times V \to V$ in coordinates as follows

$$Pad_j^i(A,T)(k_1,\ldots,k_d) = \begin{cases} A(k_1,\ldots,k_{i-1},k_{i+1},\ldots k_d) & k_i = j \\ 0 & \text{otherwise} \end{cases}$$

Equipped with this method of padding one can now define slicing along one particular mode.

Definition 4.2.6 (Tensor Family Generated by Slices along one mode)**.** Let $d > 0$ and $1 \leq i \leq d$. Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be an order d tensor, and $\mathcal{M} := \{M_1,\ldots,M_N\} \subset V_{\hat{i}}$ be a finite collection of order $d-1$ slices of the specified format along the i-th mode of T.

Define $Mod^i(T,\bar{\mathcal{M}}) : V \times V_1 \otimes \cdots \otimes V_{i-1} \otimes \mathbb{F}^N \otimes V_{i+1} \otimes \cdots V_d \to 2^V$ in terms of coordinates as follows (2^P denoting the power set of a set P)

$$Mod^i(T,\bar{\mathcal{M}}) = \left\{ T + \sum_{k=1}^{n_i} \sum_{j=1}^{N} \lambda_{j,k} Pad_k^i(S_j^i(\bar{\mathcal{M}})) | \lambda_{j,k} \in \mathbb{F} \right\}$$

Here slices are first selected from the constructed order d tensor, then padded appropriately to remain of the same format as T, then scaled and added to T.

This definition can now be extended to performing this operation along all the d modes. The only difference now notationally is writing $Mod^j(Mod^i(T, \bar{\mathcal{M}}_i), \bar{\mathcal{M}}_j)$ to denote the $Mod^j(S, \bar{\mathcal{M}}_j)$ for every $S \in Mod^i(T, \mathcal{M}_i)$ and then taking the union of all the resulting sets. Thus the final family $F := T \bmod (\mathcal{M}_1, \ldots \mathcal{M}_d)$ is a result of the d-fold composition of the distinct Mod^i operations.

Proposition 4.2.1. $Mod^j(Mod^i(T, \bar{\mathcal{M}}_i), \bar{\mathcal{M}}_j) = Mod^i(Mod^j(T, \bar{\mathcal{M}}_j), \bar{\mathcal{M}}_i)$

Proof.

$$\begin{aligned}
& Mod^j(Mod^i(T, \bar{\mathcal{M}}_i), \bar{\mathcal{M}}_j) \\
= \; & \left\{ Mod^j(S, \bar{\mathcal{M}}_j) | S \in Mod^i(T, \mathcal{M}_i) \right\} \\
= \; & \left\{ S + \sum_{a=1}^{n_j} \sum_{b=1}^{N_j} \lambda_{a,b} Pad_a^j(S_b^j(\bar{\mathcal{M}}_j)) | \lambda_{a,b} \in \mathbb{F}, S \in Mod^i(T, \mathcal{M}_i) \right\} \\
= \; & \left\{ T + \sum_{c=1}^{n_i} \sum_{d=1}^{N_i} \lambda_{c,d} Pad_c^i(S_d^i(\bar{\mathcal{M}}_i)) + \sum_{b=1}^{n_j} \sum_{a=1}^{N_j} \lambda_{a,b} Pad_a^j(S_b^j(\bar{\mathcal{M}}_j)) | \lambda_{a,b}, \lambda_{c,d} \in \mathbb{F} \right\} \\
= \; & \left\{ S + \sum_{c=1}^{n_i} \sum_{d=1}^{N_i} \lambda_{c,d} Pad_c^i(S_d^i(\bar{\mathcal{M}}_i)) | \lambda_{c,d} \in \mathbb{F}, S \in Mod^j(T, \mathcal{M}_j) \right\} \\
= \; & \left\{ Mod^i(S, \bar{\mathcal{M}}_i) | S \in Mod^j(T, \mathcal{M}_j) \right\} \\
= \; & Mod^i(Mod^j(T, \bar{\mathcal{M}}_j), \bar{\mathcal{M}}_i)
\end{aligned}$$

□

The above proof shows that the order of applying modulo operations on a given tensor does not affect the final resultant linear span.

Observation. Here, the family $F \subseteq V_1 \otimes \cdots \otimes V_d$, i.e. all its elements remain of the same order as the original tensor T. Figure 7 illustrates how one one such element from such a family can be generated.

Remark. As with the adjoining operation for symmetric tensors, generating a family of symmetric tensors modulo symmetric slices also follows immediately from the definitions provided above, but in this case it is the same family of order $d-1$ slices being used along every mode.

Since performing the modulo operation on a tensor gives a family instead of a single output tensor, one can talk about how the maximal or minimal rank of the tensor is preserved or not preserved.

Combining Adjoining and Modulo

As seen in the previous two subsections, rank information of a tensor does change when adjoining slices, and the following theorem is becomes important in Wang and Seigal's 2023 paper .

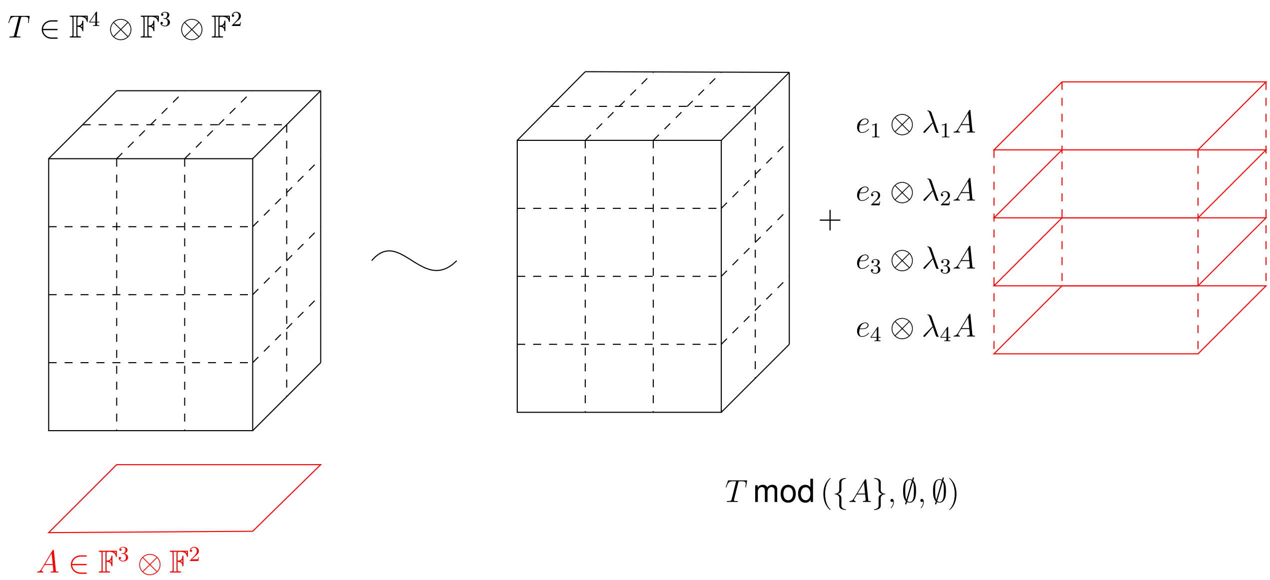

Figure 7: Family generated from a tensor T by a slice A

Theorem 4.2.1 ([27]). *Suppose $T \in V := V_1 \otimes \cdots \otimes V_d$ is an order d tensor, and $\mathcal{M}_i \subset V_{\hat{i}}$ for every $1 \leq i \leq d$. Then*

$$\textit{rank}\,(Adj(T, \mathcal{M}_1, \ldots, \mathcal{M}_d)) \geq \min \textit{rank}\,(T \textit{ mod}\,(\mathcal{M}_1, \ldots, \mathcal{M}_d)) + \sum_{j=1}^{d} \dim span(\mathcal{M}_j)$$

Equality holds if each of the $\mathcal{M}_i$ consist of rank-one tensors.

4.2.4 Cloning Tensors

The idea of cloning a tensor arises from necessity, because of difficulties that arise in the construction of a counterexample by Shitov, in the construction of an emulating family. Cloning a tensor, as the name suggests, seeks to preserve the properties of the original, while embedding it in a much "larger space".

Definition 4.2.7 (Cloning a Tensor)**.** Fix $d > 0$. Let $T \in V := V_1 \otimes \cdots \otimes V_d$ be an order d-tensor.

The n-th cloning function $C_n : V \to (\bigoplus_{i=1}^{n} V_1) \otimes \cdots \otimes (\bigoplus_{i=1}^{n} V_d)$ can be defined in coordinates in the following manner

$$C_n(T)(k_{1,j_1}, \ldots, k_{d,j_d}) = T(k_1, \ldots, k_d)$$

here, the notation $k_{i,j_i} := k_{(j_i-1)n+i}$. where $1 \leq j_i \leq n$.

Observation. Since the definition of cloning depends on a non-zero d, one can clone tensors of any order, in particular vectors, as well. This leads to the description of the clone of a standard ordered basis vector $e_i \in V$, an $\mathbb{F}$ vector space. $C_n(e_i) = \sum_{j=1}^{n} e_{i,j}$

Lemma 4.2.2. *The cloning operation as defined above distributes over the tensor product i.e. $C_n(A \otimes B) = C_n(A) \otimes C_n(B)$ where A, B are vectors in an arbitrary vector space.*

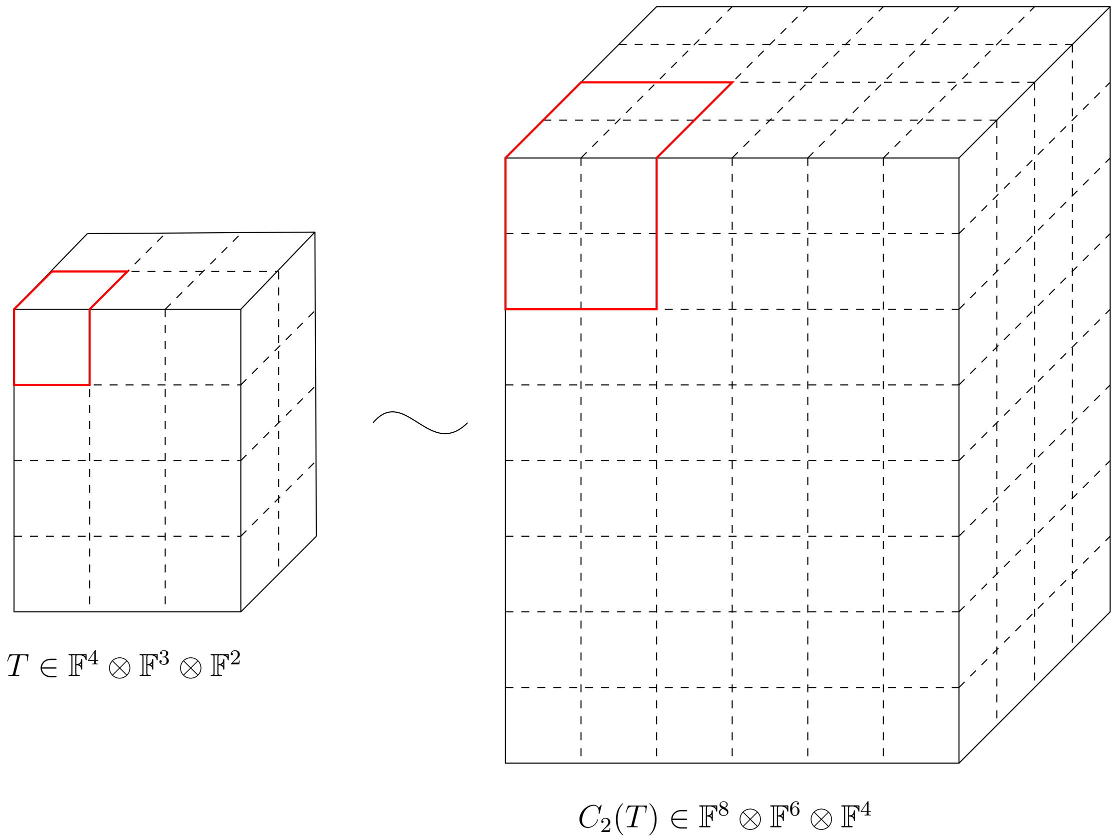

Figure 8: 2 clone of a tensor T

Proof. Consider two arbitrary vectors, $A := (a_1, \dots, a_k), B := (b_1, \dots, b_k) \in V$, an $\mathbb{F}$ vector space. Then $A \otimes B$ is an order 2 tensor (a matrix) lying in $V \otimes V$, and $(A \otimes B)(i, j) = a_i b_j$ Then according to definition 4.2.7,

$$C_n(A \otimes B)(k_1, k_2) = (A \otimes B)(k_1(\bmod n), k_2(\bmod n)) = A(k_1(\bmod n))B(k_2(\bmod n)),$$

here indicating the regular modulo operation from number theory. This tensor lies in $(\bigoplus_{i=1}^n V) \otimes (\bigoplus_{i=1}^n V)$.

Now taking the vectors $C_n(A), C_n(B) \in (\bigoplus_{i=1}^n V)$. Here

$$C_n(A) = (\underbrace{a_1, \dots, a_1}_{n \text{ times}} \dots \underbrace{a_k, \dots, a_k}_{n \text{ times}}),$$

and likewise for $C_n(B)$. Their tensor product is also a matrix lying in $(\bigoplus_{i=1}^n V) \otimes (\bigoplus_{i=1}^n V)$ where now

$$(C_n(A) \otimes C_n(B))(k_1, k_2) = C_n(A)(k_1)C_n(B)(k_2) = A(k_1(\bmod n))B(k_2(\bmod n)).$$

Thus, since all the components of each tensor is equal, they are the same tensor. $\square$

Remark. This property, along with the observation that cloning is a linear operation, allows for the proof that cloning a tensor does preserve its rank.

Another description of cloning is by first transforming the original by tensor product. Denote $\mathbf{1}_n$ as the vector of 1's lying in $\mathbb{F}^n$. Then the n-clone of a simple vector v can be described as $(v \otimes \mathbf{1}_n)^{\emptyset}$ or the flattening of the matrix created by tensoring v with the vector of ones. Thus another way to look at $C_n(T)$ is with the following procedure:

$$\begin{aligned}
C_n(T) &= C_n\left(\sum_{i=1}^{r} T_i\right) \\
&= \sum_{i=1}^{r} C_n(T_i) \\
&= \sum_{i=1}^{r} C_n(v_1^i \otimes \cdots \otimes v_d^i) \\
&= \sum_{i=1}^{r} C_n(v_i^i) \otimes \cdots \otimes C_n(v_d^i) \\
&= \sum_{i=1}^{r} (v_1^i \otimes \mathbf{1}_n)^{\emptyset} \otimes \cdots \otimes (v_d^i \otimes \mathbf{1}_n)^{\emptyset}
\end{aligned}$$

Proposition 4.2.2. *rank* $(T) =$ *rank* $(C_n(T))$ *for* C_n *the cloning operation as defined in definition 4.2.7, and* $T \in V := V_1 \otimes \cdots \otimes V_d$*.*

Proof. The proof is straightforward, and is as follows

$\leq$**:** Since T can be seen as a sub-tensor lying in $C_n(T)$, this inequality naturally exists. The rank of a subtensor is always at most the rank of the tensor, but never more.

$\geq$**:** Since cloning distributes over addition (lemma 4.2.2), if one has the rank decomposition $T = \sum_{i=1}^{r} T_i$, then $C_n(T) = \sum_{i=1}^{r} C_n(T_i)$ is a decomposition of $C_n(T)$. Thus the rank of $C_n(T)$ is at most that of T.

□

4.3 The Counterexample over $\mathbb{R}$

Wang and Seigal [27] in 2023 published a paper that demonstrated the construction of a distinct counterexample to Comon's conjecture, following Shitov [23]. Their paper first examined the constructions made by Shitov, established new notation and terminology to prove results on rank pre- and post-operations, and finally they used MATLAB to construct a distinct order 6 example with real rank 30913 and real symmetric rank 30914.

As the title of the paper suggests, such a counterexample relies on certain lower bounds and inequalities remaining strict, particularly when making constructions in a field that is not algebraically complete.

4.3.1 Details of Construction

The paper in 2023 [27] works over $\mathbb{R}$, specifically in $\mathbb{R}[x, y]$ as the underlying ring of polynomials from which the homogenous polynomials can be selected. For simplicity's sake, there are only 2 variables, corresponding to an underlying real vector space of dimension 2.

The particular tensor under consideration is represented by the homogenous polynomial $x^d - ky^d$ where $d > 0$ is a fixed, even integer, and k (also fixed) is any non-zero positive constant. Such a homogenous polynomial corresponds to the following tensor

$$T := (e_1)^{\otimes d} - k(e_2)^{\otimes d}, k > 0, d = 2n, n \in \mathbb{Z}_+ \tag{4.1}$$

Visualised as a multiway array, this is a tensor that has non-zero entries at the two extreme "corners" and 0's everywhere else. Such a tensor clearly has rank and symmetric rank as 2.

Since the stated goal is to construct a counterexample to Comon's conjecture, this initial "seed" tensor is first modified by the substitution method (section 4.2.3) to generate a linear span of tensors. Such a family when constructed has elements having minimal rank and symmetric rank. The elements in this family achieving these minimal values do not have to be the same.

The family chosen in this case is

$$\mathcal{M} := \{x^{d-1-i}y^i\}_{i=1}^{d-2} \tag{4.2}$$

This is a collection of monomials of order $d-1$ that are of "mixed" form i.e. they are monomials not purely in either variable. When describing such a family in terms of symmetric tensors (see section 2.3), the family can be written as

$$\mathcal{M} := \left\{ \frac{1}{(d-1)!} \sum_{\sigma \in S_{d-1}} \sigma((e_1)^{\otimes d-1-i} \otimes (e_2)^{\otimes i}) \right\}_{i=1}^{d-2} \tag{4.3}$$

These order $d-1$ tensors, since not formed of purely only one standard ordered basis vector, do not affect the initial tensor T, when performing the operation of generating a family modulo slices: $T \bmod \mathcal{M}$. (Note here since they are working with symmetric tensors, it is enough to specify only one family.)

Establishing A First Difference in Rank and Symmetric Rank

Within the constructed family $T \bmod \mathcal{M}$, a tensor $P \in T \bmod \mathcal{M}$ corresponds to the following polynomial

$$P(x,y) = x^d - ky^d + \sum_{i=1}^{d-2} (\alpha_i x + \beta_i y) x^{d-1-i} y^i, \alpha_i, \beta_i \in \mathbb{R} \forall i = 1, \ldots, d-2, k > 0 \tag{4.4}$$

First, the symmetric rank decomposition is computed, and shown to be at least 2 or higher.

Proposition 4.3.1 ([27]). *A polynomial of the form $P(x,y)$ (as eq. (4.4)) has symmetric rank at least 2.*

Proof. Suppose not. Then one can rewrite $P(x,y)$ in terms of a linear polynomial raised to the dth power i.e.

$$P(x,y) = (ax + by)^d$$

for some choices of $a, b \in \mathbb{R}$.
Expanding the RHS would then lead to the following

$$\begin{aligned} P(x,y) &= (ax+by)^d \\ &= \sum_{j=0}^{d} \binom{d}{j} a^{d-j} b^j x^{d-j} y^j \end{aligned}$$

leading to the following equality of polynomial expressions

$$x^d - ky^d + \sum_{i=1}^{d-2} (\alpha_i x + \beta_i y) x^{d-1-i} y^i = \sum_{j=0}^{d} \binom{d}{j} a^{d-j} b^j x^{d-j} y^j$$

This leads to a contradiction simply by looking at the coefficient of the y^d monomial on either side of the equality. One must have $b^d = -k$ for some choice of $b \in \mathbb{R}$. Since such an equation has no real solutions, the initial assumption of being able to write P in terms of the d-th power of a single linear form was false, and therefore the Waring rank (equivalent to the symmetric rank) of $P \geq 2$. $\square$

This establishes $\min_{P \in T \bmod \mathcal{M}} \operatorname{srank}(P) = 2$. The next step in the paper is the establishing of the fact that the minimum *rank* element of such a family has rank 1.

Proposition 4.3.2 ([27]). *There is a tensor of the form $P(x,y)$ (see eq. (4.4)) that is an elementary tensor*

Proof. Consider the original tensor $T := x^d - ky^d$ (see eq. (4.1)). This is a tensor of the form eq. (4.4). Now summing all the elements in $\mathcal{M}$ as defined earlier (see eq. (4.2)) gives an order $d-1$ tensor with non-zero entries everywhere except for two "corners" — corresponding to x^{d-1} and y^{d-1}. These can be added along the d-th mode to the first slice, and after scaling by a factor of $-k$, to the last slice.

Formally the tensor T is now modified to

$$x^d - ky^d + \underbrace{\sum_{i=1}^{d-2} x^{d-i} y^i}_{\text{modifying the first } d\text{-slice}} + \underbrace{-k\left(\sum_{j=1}^{d-2} x^{d-1-j} y^{j+1}\right)}_{\text{scaling and modifying the last } d\text{-slice}}$$

An important point to note here is that changing the degree of the polynomial changes the order, and since the tensors all lie in $(\mathbb{R}^2)^{\otimes d}$, changes affecting the "first" slice have an extra factor of x, and changes affecting the "last" or second have an extra factor of y.

The final step of construction is adding the order $d-1$ tensor corresponding to the monomial $x^{d-2}y$ (scaled by an appropriate a_i) to the first i-slice for all d modes of the tensor T. Likewise for the tensor corresponding to the monomial xy^{d-2} (with appropriate b_i added to the last i-slices). The choices of these scaling coefficients are such that

$$\left(\sum_{i=1}^{d} a_i\right) - a_j = 0, \forall j \neq d \tag{4.5}$$

$$\sum_{i=1}^{d-1} a_i = -k \tag{4.6}$$

$$\left(\sum_{i=1}^{d} b_i\right) - b_j = 0, \forall j \neq d \tag{4.7}$$

$$\sum_{i=1}^{d-1} b_i = 1 \tag{4.8}$$

These choices for a_i are made because modifying the tensor by adding the scaled tensors modifies only one entry in last d slice. The last addition of $x^{d-1}y$ along the d-mode only affects the first d slice. A similar line of reasoning follows the choices for the b_i. Making such choices is always possible, and non-unique.

Performing these operations results in a tensor still lying in the family $T \bmod \mathcal{M}$ now of the form $(\mathbf{1}_2)^{\otimes d-1} \otimes (1, -k)$, an elementary order d tensor. Here again $\mathbf{1}_n$ denotes the vector of all 1's lying in $\mathbb{F}^n$ □

Remark. It is useful to note that proving such an elementary tensor exists is not a function of the underlying field. (assuming characteristic 0). Such a proof could also be modified for any number of variables (moving from working in $\mathbb{F}^2$ to $\mathbb{F}^n$), and for tensors of any even order d.

The following statement holds for the family $T \bmod \mathcal{M}$

$$\min \text{rank}\ (T \bmod \mathcal{M}) < \min \text{srank}\ (T \bmod \mathcal{M}) \tag{4.9}$$

Moving from Families to a Specific Tensor

Now that the existence of a family having elements with differing minimal rank and symmetric rank has been established, the authors construct $SAdj(T, \mathcal{M})$, as a symmetric tensor that is a candidate counterexample to Comon's Conjecture. The problem of computing the rank of such a construction now arises, as it relates to the original T and candidate family $\mathcal{M}$.

The solution proposed originally by Shitov [23], is a proposed modification of $\mathcal{M}$ to a family of *elementary* order $d-1$ tensors that span the same linear space. Unfortunately, in the current format, doing so also violates the vital inequality required for rank and symmetric rank of the family $T \bmod \mathcal{M}$ (see eq. (4.9))

Such a modification is required for the original computation of rank (see theorem 4.2.1) to be an equality. The following proposition is adapted from [27], to show that modifying $\mathcal{M}$ in such a direct way violates eq. (4.9). It is enough to show that an elementary symmetric tensor lies in the linear space generated with the modified family.

Proposition 4.3.3. *Consider T as in eq. (4.1), and $\mathcal{M}$ as eq. (4.2). Let $\mathcal{W} \subset (\mathbb{R}^2)^{\otimes d-1}$, a finite set of elementary symmetric tensors such that $\mathcal{W}$ spans $\mathcal{M}$. Then $0 \in T \bmod \mathcal{W}$*

Proof. Consider $\mathcal{W}$, a collection of elementary symmetric tensors of appropriate order. Showing $0 \in T \operatorname{mod} \mathcal{W}$ is equivalent to asking that $\mathcal{W}$ spans the set $\{x^i y^d - i\}_{i=0}^{d-1} \supseteq \mathcal{M}$ This is due to the fact that every slice of T (seen as an order $d-1$ tensor and therefore represented by a degree $d-1$ polynomial) can then be expressed as lying in the span of $\mathcal{W}$. In particular, if one can show that the dimension of span $\mathcal{W}$ is d (the number of degree $d-1$ monomials in 2 variables), this completes the proof.

Suppose the dimension of the span of $\mathcal{W}$ is less than d. Then consider the tensor represented by $x^{d-2}y$. This monomial has rank $d-1$ since it can be written minimally as sum of $d-1$ elementary tensors. Thus the dimension of span $\mathcal{W}$ is $d-1$. Since $\mathcal{W}$ consists of symmetric tensors, its basis vectors must be a linear forms raised to the $d-1$th power: $(\lambda_i x + \mu_i y)^{d-1}$. These forms must also correspond to elementary tensors, by virtue of being elements of $\mathcal{W}$.

Now since this set, $\{(\lambda_i x + \mu_i y)^{d-1}\}_{i=1}^{d-1}$ is a basis for span $\mathcal{W}$, one must have that the tensors $x^{d-2}y$ and xy^{d-2} can be written as linear combinations of the elements of the set. Doing so imposes conditions on $\lambda_i, \mu_i > 0$. These conditions are as follows: $\square$

This failure of eq. (4.9) after modifying $\mathcal{M}$ to only contain elementary symmetric tensors is why the need for cloning arises. As demonstrated in section 4.2.4, since cloning preserves rank, then inequality is preserved after taking the k-th clones, and transforms into

$$\min \text{rank } C_k(T) \operatorname{mod} C_k(\mathcal{M}) < \min \text{srank } C_k(T) \operatorname{mod} C_k(\mathcal{M}) \tag{4.10}$$

In this situation, modifying $C_k(M)$ - a family of order $d-1$ tensors lying in $C_k(V^{\otimes d-1})$ - into $\mathcal{W}$ such that the above modification procedure can occur successfully. This results in working with a symmetric tensor $Sadj(C_k(T), \mathcal{W})$ - the final symmetric tensor having a definite distinction between rank and symmetric rank.

4.3.2 Emulating Family Construction

Chapter 5

Conclusion

5.1 Further Work

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